



USER MANUAL

Canvas Assignment Downloader & AI Evaluator

A Step-by-Step Guide for Faculty (Beginner Friendly)

Scripts Covered:

1. Assignment-Downloader.py
2. evaluate_assignments-groq.py

1. What Do These Scripts Do?

You have two Python scripts that work together as a team:

Script	What it does
Assignment-Downloader.py	Connects to your Canvas course, downloads every student's submitted file, packs them into a ZIP, and extracts them into a submissions/ folder on your computer.
evaluate_assignments-groq.py	Reads each student's file from the submissions/ folder, sends it to the Groq AI model, gets rubric-based marks and comments back, uploads the grades to Canvas, and saves a CSV summary.

⚡ Always run Script 1 first, then Script 2. Script 2 needs the submissions/ folder created by Script 1.

2. One-Time Setup (Do This Once)

2.1 Install Python

Make sure Python 3.8 or newer is installed on your computer.

1. Open your browser and go to: <https://www.python.org/downloads/>
2. Download the latest version and run the installer.
3. On Windows, tick "Add Python to PATH" before clicking Install.
4. Open a terminal / command prompt and verify:

```
python --version
```

2.2 Install Required Libraries

Open a terminal and run this single command:

```
pip install requests groq pandas python-docx pymupdf pillow
```

💡 If pip is not found, try: `python -m pip install requests groq pandas python-docx pymupdf pillow`

2.3 Save the Scripts

Create a folder anywhere on your computer, for example:

`C:\Faculty\Grading\` (Windows) or `~/Faculty/Grading/` (Mac/Linux)

Save both .py files into that same folder.

3. Getting Your API Keys

Both scripts need two API keys: one from Canvas and one from Groq. Think of them as passwords that prove you are allowed to access those services.

3.1 Canvas API Token

What is it?

A secret code that lets the scripts talk to your Canvas account on your behalf.

How to generate it — step by step:

5. Open Canvas in your browser and log in.
6. Click your profile picture (top-right corner) → click "Account".
7. Click "Settings" from the left-hand menu.
8. Scroll down to the section called "Approved Integrations" and click "+ New Access Token".
9. In the "Purpose" box type something like: Assignment Grader Script
10. Leave "Expires" blank (or set a date if your institution requires it).
11. Click "Generate Token".
12. Copy the long string shown — it starts with 7~ or similar.

 **IMPORTANT:** Canvas will only show the token *ONCE*. Copy it immediately and paste it in a safe place (like Notepad). You cannot see it again after closing that dialog.

Where to paste it in the scripts:

In BOTH scripts, find this line and replace the value:

`CANVAS_API_TOKEN = "7~th3yzNkAGFk..."` ← replace everything inside the quotes

3.2 Groq API Key

What is it?

A key that lets the evaluator script use the Groq AI service (which runs the Llama AI model) to read and grade student submissions.

How to generate it — step by step:

13. Open your browser and go to: <https://console.groq.com>
14. Click "Sign Up" (free) or log in if you already have an account.
15. After logging in, click "API Keys" in the left sidebar.
16. Click "+ Create API Key".
17. Give it a name like: Assignment Grader
18. Copy the key shown — it starts with gsk_

 **IMPORTANT:** Groq also shows the key only once. Save it immediately.

Where to paste it in the scripts:

In Script 1 (Assignment-Downloader.py):

```
GROQ_API_KEY = "gsk_..." ← paste your Groq key here
```

In Script 2 (evaluate_assignments-groq.py):

```
GROQ_API_KEY = "gsk_..." ← paste your Groq key here
```

4. Finding Your Canvas Course ID & Assignment ID

Every Canvas course and assignment has a unique number in its web address (URL). Here is how to find them.

4.1 Course ID

19. Open Canvas and go to your course.
20. Look at the browser address bar. The URL looks like:
<https://canvas.instructure.com/courses/14222223>
21. The number at the end is your Course ID. In this example: 14222223

4.2 Assignment ID

22. Inside your course, click on Assignments.
23. Click the specific assignment you want to download/grade.
24. Look at the URL:
<https://canvas.instructure.com/courses/14222223/assignments/62461759>
25. The last number is your Assignment ID. In this example: 62461759

4.3 Canvas URL

Your institution may use a custom Canvas domain. Check your browser's address bar when logged into Canvas:

- If it says `canvas.instructure.com` → use that (already the default in the scripts)
 - If it says something like `lms.youruniversity.edu` → use that URL instead
-

5. Script 1 — Assignment-Downloader.py: Full Configuration Guide

Open Assignment-Downloader.py in any text editor (Notepad, VS Code, etc.) and change these values at the top of the file under the CONFIGURATION section:

Variable / Line	What to Change	Example / Notes
GROQ_API_KEY	Your Groq key starting with gsk_	gsk_abc123...
CANVAS_API_TOKEN	Your Canvas token starting with 7~	7~xyz789...
CANVAS_URL	Your Canvas domain	"https://canvas.instructure.com"
COURSE_ID	Your course number from the URL	"14222223"
ASSIGNMENT_ID	Your assignment number from the URL	"62461759"
SUBMISSIONS_FOLDER	Folder name where files are saved (keep as default)	"submissions"
ZIP_FILE	Name of the ZIP file created (keep as default)	"submissions.zip"
MAX_WORKERS	Number of parallel downloads (10 is fine for most)	10

💡 All values must be inside double-quotes " " because they are text strings in Python.

6. Script 2 — evaluate_assignments-groq.py: Full Configuration Guide

Open evaluate_assignments-groq.py and update these settings:

Variable / Line	What to Change	Example / Notes
GROQ_API_KEY	Your Groq key (same as Script 1)	gsk_abc123...
CANVAS_API_TOKEN	Your Canvas token (same as Script 1)	7~xyz789...
CANVAS_URL	Your Canvas domain (same as Script 1)	"https://canvas.instructure.com"
COURSE_ID	Same course ID as Script 1	"14222223"
ASSIGNMENT_ID	Same assignment ID as Script 1	"62461759"
GROQ_MODEL	AI model to use — llama-3.1-8b-instant is fast & free	"llama-3.1-8b-instant"
SUBMISSIONS_FOLDER	Must match Script 1's folder name	"submissions"
FACULTY_COMMENTS_ENABLED	Set True to type comments per student manually	False

6.1 Changing the AI Model

The default model `llama-3.1-8b-instant` is fast and free. If you want better accuracy you can use:

- `llama-3.3-70b-versatile` — smarter but slower (already in the code, just uncomment it)

To switch, change the line:

```
#GROQ_MODEL = "llama-3.3-70b-versatile" ← remove the # to enable this  
GROQ_MODEL = "llama-3.1-8b-instant" ← add a # to disable this one
```

6.2 Faculty Comments

If `FACULTY_COMMENTS_ENABLED` is set to `True`, the script will pause after evaluation and ask you to type a personal note for each student. These notes get uploaded to Canvas along with the grade. Set it to `False` to skip this step.

7. Running the Scripts — Step by Step

Step 1 — Open a Terminal in Your Scripts Folder

- Windows: Open File Explorer, go to your scripts folder, hold Shift + right-click → "Open PowerShell window here"
- Mac: Open Terminal, type `cd` then drag your folder into the Terminal window, press Enter
- Linux: Right-click in folder → "Open Terminal Here"

Step 2 — Run Script 1 (Download Submissions)

```
python Assignment-Downloader.py
```

You will see progress messages like:

```
🔗 Fetching submissions from Canvas...  
✅ Found 35 submissions  
⚡ Downloading 35 files in parallel...  
✅ Ahmed Khan: Ahmed_Khan.pdf (124.3 KB)  
📦 Creating submissions.zip...  
📁 Extracting to submissions/...
```

When it finishes, a `submissions/` folder will appear in your scripts directory containing all student files.

✅ *Success check: Open the `submissions/` folder. You should see one file per student named like: `Priya_Sharma.pdf`*

Step 3 — Run Script 2 (Evaluate & Upload Grades)

```
python evaluate_assignments-groq.py
```

You will see:

-  Fetching assignment from Canvas...
-  Assignment : Lab 3 – Data Structures
-  Found 35 students
-  Evaluating: Ahmed Khan (1 file: Ahmed_Khan.pdf)...
-  Generating results CSV...
-  Uploading grades to Canvas...
-  Ahmed Khan: 78/100 (B)

When done, a CSV file like Lab_3_results.csv is saved in your folder with all results.

 *Success check: Open Canvas → Grades. You should see marks and rubric scores filled in for all students.*

8. Every Time You Use It for a New Assignment

You only need to change a few things each time. Here is the quick checklist:

#	What to update	In which script(s)
1	ASSIGNMENT_ID ← new assignment number from URL	Both scripts
2	COURSE_ID ← only if switching to a different course	Both scripts
3	Delete the old submissions/ folder and submissions.zip before re-running Script 1	—
4	API keys and Canvas URL stay the same — no need to change	—

9. Understanding the Output CSV

After Script 2 runs, a CSV file is saved in your folder. Open it in Excel or Google Sheets. It contains:

Column Name	Meaning
Student Name	Student's name as it appears in Canvas
Files	Filename(s) submitted by the student
[Criterion] (/max)	Score for each rubric criterion (e.g. Correctness /40)
[Criterion] Rating	Band label: Excellent / Average / Need Improvement
Total	Sum of all criterion scores

Percentage	Total as a percentage
Grade	A (≥ 85) / B (≥ 70) / C (≥ 55) / D (≥ 40) / F (< 40)
Overall Feedback	AI-generated 3–4 line comment specific to that student
Strengths	What the student did well
Improvements	Areas needing improvement
Faculty Comment	Your personal note (if FACULTY_COMMENTS_ENABLED = True)

10. Common Errors & Fixes

Error Message	What it means	How to fix it
✗ Failed to fetch submissions: 401	Canvas token is wrong or expired	Generate a new Canvas API token and paste it
✗ Failed to fetch submissions: 404	Wrong COURSE_ID or ASSIGNMENT_ID	Double-check the IDs from the Canvas URL
ModuleNotFoundError: No module named 'groq'	Libraries not installed	Run: <code>pip install groq requests pandas python-docx pymupdf pillow</code>
✗ submissions/ folder not found!	Script 1 was not run first	Run Assignment-Downloader.py first
✗ JSON parse error for [student]	AI returned an unexpected response (rare)	Re-run Script 2; it will retry automatically
⚠ [Student]: Not matched in Canvas	Student name in file doesn't match Canvas	Manually upload the grade in Canvas Gradebook
AuthenticationError: Invalid API Key (Groq)	Groq key is wrong or expired	Generate a new Groq API key at console.groq.com

11. Quick Reference Card

Cut out or screenshot this section for easy reference:

🔑 Where to get API keys	📄 Configuration variables to change per assignment
Canvas Token: Canvas → Account → Settings → New Access Token Groq Key: console.groq.com → API Keys → Create	• ASSIGNMENT_ID (every assignment) • COURSE_ID (when switching courses) • CANVAS_API_TOKEN (when it expires) • GROQ_API_KEY (when it expires)

▶ Order to run scripts	📁 Output files created
1. python Assignment-Downloader.py 2. python evaluate_assignments-groq.py	• submissions/ (folder with all student files) • - submissions.zip (backup zip) • [Assignment_Name]_results.csv (grades + feedback)

💡 *Keep this manual handy for the first few times. After 2–3 runs you will remember the steps by heart!*